1. COMPUTING WITH REAL NUMBERS

Floating-point system

Normalized binary FPN: $\mathrm{fl}(x)=\pm(1.d_1d_2\ldots d_p)_2\,2^e$, $d_j\in\{0,1\}$, $e_{\min}\le e\le e_{\max}$. IEEE double: sign 1 bit, p=52 fraction bits, 11 exponent bits; normal $e\in[-1022,1023]$.

$\varepsilon=2^{-52}\approx 2.22\times10^{-16}$ (gap from 1 to next FPN);

$\varepsilon_i=2^m\varepsilon\ (2^m\le i<2^{m+1}),\quad x_{\max}=(2-\varepsilon)2^{1023},\quad x_{\min}^{\text{normal}}=2^{-1022}$. Below: subnormal, then underflow to 0; overflow→±∞; undefined→NaN. ***FPN arithmetic is not associative; avoid exact equality tests.***

Errors, accuracy, stability

For approximation $\bar{x}$ to x: $e_a=|\bar{x}-x|,\quad e_r=\frac{|\bar{x}-x|}{|x|}$ (***x≠0***).

Significant decimal figures: largest p satisfying $e_r<\frac{1}{2}10^{-p}$.

Storage rounding error $\le\varepsilon|x|$.

Catastrophic cancellation: subtracting computed nonzero $\bar{x},\bar{y}$ when $x\approx y$ can greatly amplify relative error.

Quadratic $ax^2+bx+c=0$:

$q=-\frac{1}{2}\left(b+\mathrm{sign}(b)\sqrt{b^2-4ac}\right),\quad x_1=\frac{q}{a},\quad x_2=\frac{c}{q}$. ***Require a≠0, real-root form $b^2-4ac\ge0$, q≠0 for c/q.***

Cost and storage

1 bit∈{0,1}; 1 byte=8 bits; KiB=2^{10} bytes, MiB=2^{20}, GiB=2^{30}. Signed b-bit integer range $[-2^{b-1},2^{b-1}-1]$. 1 flop = one FPN +, −, ×, /.

Speed=(cycles/s)(cores)(flops/cycle); time=flops/speed.

$\mathrm{memory}(n\times n)=8n^2$ bytes, $\quad\mathrm{flops}(AB)=n^2(2n-1)\approx2n^3$.

2. NORMS, EQUALITY, SENSITIVITY

Vector norms and error

For $x\in\mathbb{R}^n$, p≥1: $\|x\|_p=\left(\sum_{i=1}^n|x_i|^p\right)^{1/p}$;

$\|x\|_1=\sum_i|x_i|,\quad \|x\|_2=\sqrt{x^Tx},\quad \|x\|_\infty=\max_i|x_i|$. Norm axioms: positivity/definiteness, triangle inequality, absolute homogeneity. $x=y\iff\|x-y\|=0,\quad \|x-y\|\le\tau(\varepsilon,n,\|x\|,\|y\|)$.

$e_a=\|\bar{x}-x\|,\quad e_r=\frac{\|\bar{x}-x\|}{\|x\|}$ (***x≠0***);

$\frac{\|\bar{x}-x\|_\infty}{\|x\|_\infty}<\frac{1}{2}10^{-k},\quad k=\lfloor-\log_{10}(2e_{r,\infty})\rfloor$.

Matrix norms

Subordinate norm: $\|A\|_p=\max_{x\neq0}\frac{\|Ax\|_p}{\|x\|_p}$;

$\|Ax\|_p\le\|A\|_p\|x\|_p,\quad \|AB\|_p\le\|A\|_p\|B\|_p$.

$\|A\|_1=\max_j\sum_i|a_{ij}|,\quad \|A\|_\infty=\max_i\sum_j|a_{ij}|,\quad \|A\|_2=\sqrt{\lambda_{\max}(A^TA)}=\sigma_1$;

Frobenius $\|A\|_F=\sqrt{\Sigma_{ij}|a_{ij}|^2}$ (***not generally $\|A\|_2$***).

Conditioning of Ax=b

For nonsingular A: $\kappa(A)=\|A\|\,\|A^{-1}\|\ge1$; rcond=1/κ.

$\kappa_2(A)=\frac{\max_i|\lambda_i|}{\min_i|\lambda_i|}\ (A=A^T),\quad \kappa_2(A)=\frac{\sigma_1}{\sigma_n}\ (\mathrm{rank}(A)=n)$.

$(A+\delta A)(x+\delta x)=b+\delta b,\quad \|(A+\delta A)^{-1}\|\le\alpha\|A^{-1}\|$; A+δA nonsingular, α>1. $\rho_A=\frac{\|\delta A\|}{\|A\|},\quad \rho_b=\frac{\|\delta b\|}{\|b\|}$; $\frac{\|\delta x\|}{\|x\|}\le\alpha\kappa(A)(\rho_A+\rho_b)$.

Large κ ⇒ sensitive/ill-conditioned; κ≳1/ε can destroy machine-precision information.

3. MATRIX STRUCTURE & FACTORISATIONS

Special matrices

$C_{ij}=E[(X_i-\mu_i)(X_j-\mu_j)],\quad C_{ii}=\sigma_i^2$; C symmetric PSD.

$K_{ij}=\frac{C_{ij}}{\sigma_i\sigma_j},\quad K=\Sigma^{-1}C\Sigma^{-1},\quad K_{ii}=1$, $|K_{ij}|\le1$ (***$\sigma_i,\sigma_j>0$***).

Upper/lower triangular: $a_{ij}=0$ for i>j / i<j. $\det A=\prod_i a_{ii},\quad A$ nonsingular $\iff a_{ii}\neq0\ \forall i$. Unit triangular: $a_{ii}=1$. Triangular solve ≈n² flops.

$a_{ij}=0\quad$ if $\quad i-j>m_\ell$ or $j-i>m_u$; total bandwidth $m_\ell+m_u+1$; nnz$\le(1+m_\ell+m_u)n-m_\ell(m_\ell+1)/2-m_u(m_u+1)/2$. Tridiagonal: $m_\ell=m_u=1$. Toeplitz: $a_{ij}=\alpha_{j-i}$.

Symmetric $A^T=A$; skew $A^T=-A$. $A=\frac{A+A^T}{2}+\frac{A-A^T}{2}$. Definiteness by x^TAx: PD >0, PSD ≥0, ND <0, NSD ≤0; for real symmetric A, signs of eigenvalues give type. Strict row diagonal dominance: $|a_{ii}|>\sum_{j\neq i}|a_{ij}|$; symmetric + strict dominance ⇒ PD.

$Q^TQ=I,\quad Q^{-1}=Q^T,\quad \kappa_2(Q)=1,\quad \|Qx\|_2=\|x\|_2$.

Permutation matrices are orthogonal.

LU (general square systems)

With partial pivoting $PA=LU$, L unit lower and U upper; $|\ell_{ij}|\le1$. Solve Ax=b by $Ly=Pb,\quad Ux=y$; $\det(A)=\det(P)\prod_i u_{ii}$.

Factor cost ≈2n³/3, then ≈2n² per RHS; reuse factors for many b. ***Require nonsingular A for a unique solve; do not form A^{-1}.***

Cholesky (symmetric PD systems)

$A=R^TR$, R upper. ***Require real symmetric positive-definite A.***; $r_{ii}>0$ makes R unique (otherwise DR, D=diag(±1), also factors A). Solve Ax=b by $R^Ty=b,\quad Rx=y$. $\det(A)=\prod_i r_{ii}^2>0,\quad \kappa_2(A)=\kappa_2(R)^2$. Factor cost n³/3, then 2n² per RHS; no row pivoting.

QR (least squares / rank)

For $A\in\mathbb{R}^{m\times n}$, m≥n, rank(A)=n:

$A=Q\begin{bmatrix}R\\0\end{bmatrix}=YR,\quad Q=[Y\ Z],\quad Q^TQ=I$; R nonsingular upper.

For $\min_x\|Ax-b\|_2$, $Rx=Y^Tb,\quad b-YY^Tb\perp\mathrm{col}(A)$. ***Avoid normal equations when conditioning matters.*** Cost ≈2mn²−(2/3)n³. If rank deficient: $AP=Q\begin{bmatrix}R\\0\end{bmatrix}$, column pivoting, $|r_{11}|\ge\cdots\ge|r_{nn}|$; small diagonal entries indicate numerical rank deficiency.

SVD (rank / pseudoinverse)

$A\in\mathbb{R}^{m\times n}$, m≥n: $A=U\begin{bmatrix}\Sigma\\0\end{bmatrix}V^T$; U,V orthogonal,

$\Sigma=\mathrm{diag}(\sigma_1\ge\cdots\ge\sigma_n\ge0)$. rank(A)=number of positive σ_i;

$A=\sum_{j=1}^r\sigma_ju_jv_j^T,\quad Av_i=\sigma_iu_i,\quad u_i^TA=\sigma_iv_i^T$;

$\sigma_i^2=\lambda_i(A^TA),\quad \|A\|_2=\sigma_1$. Full rank: $\kappa_2(A)=\sigma_1/\sigma_n$.

$A^+=V[\Sigma^+\ \ 0]U^T,\quad x_{\min}=A^+b$ gives least-squares x for full column rank and minimum-norm least-squares x when rank deficient; Σ^+ uses reciprocals of retained positive σ_i. Numerical rank tolerance ≈max(m,n)εσ₁.

3. MATRIX RULES & EXAM TACTICS (CONT.)

Det/eig: det(AB)=det(A)det(B), $\det(A^T)=\det(A)$, $\det(A^{-1})=1/\det(A)$, $\det(A^k)=\det(A)^k$; A singular ⇔ det(A)=0. det(A−λI)=0; triangular $\lambda_i=a_{ii}$; $\det(A)=\Pi\lambda_i$, $\mathrm{tr}(A)=\Sigma\lambda_i$; symmetric PD ⇔ $\lambda_{\min}>0$.

Gram/solve: $z^TD^TDz=\|Dz\|_2^2\ge0$; D^TD PD ⇔ rank(D)=n. $B^{-1}Ax=b$ ⇒ Ax=Bb; for $M_1\cdots M_kx=b$, solve the factors in order.

Q/P/powers/error: A=QR=(QS)(SR), S=diag(±1); PA permutes rows, AP columns. M^n: binary repeated squaring O(log n). $\rho_A=0$: $\rho_x\lesssim\kappa\rho_b$; $\rho_A\approx\rho_b\approx\varepsilon$: $\rho_x\lesssim2\kappa\varepsilon$.

4. PARAMETRIC DATA FITTING

Residual model

Data (t_i,y_i), i=1,…,m; parameters $x\in\mathbb{R}^n$, ***m≥n***.

$r_i(x)=\phi(x;t_i)-y_i,\quad x^*=\arg\min_x\|r(x)\|_p$. Interpolation ⇔ r=0 (***appropriate only for highly accurate data***).

Linear-in-parameters:

$\phi(x;t)=\sum_jx_j\phi_j(t),\quad A_{ij}=\phi_j(t_i),\quad r=Ax-y$.

Linear least squares (p=2)

Normal equations: $A^TAx^*=A^Ty$; residual orthogonal to col(A). A^TA PSD, and PD iff A has full column rank. ***Avoid forming A^TA when conditioning matters. $\kappa_2(A^TA)=\kappa_2(A)^2$.***

Pivoted QR AP=Q[R;0], Q=[Y Z]: $R(P^Tx^*)=Y^Ty$. SVD $A=U[\Sigma;0]V^T$, U=[Y Z]: $x^*=V\Sigma^{-1}Y^Ty,\quad x_{\min}^*=V\Sigma^+Y^Ty$. Use $\Sigma^+=\mathrm{diag}(1/\sigma_1,\ldots,1/\sigma_r,0,\ldots)$ for rank r<n.

Choice of norm / transformation

p=2: suited to independent N(0,σ²) errors with common variance. p=1: robust to outliers, $\min\sum_iv_i:\quad -v_i\le r_i\le v_i,\quad v_i\ge0$. p=∞: $\min v:\quad -v\le r_i\le v,\quad v\ge0$.

Linear r gives linear programs.

Log-linearisation: products/exponentials/powers may be transformed. ***Log transform requires every logged quantity positive and any shifted base $t-\beta_j>0$.***

5. CUBIC SPLINES & SPARSE MATRICES

Polynomials and interpolation

$p(t)=\sum_{k=0}^na_kt^k=a_0+t(a_1+t(a_2+\cdots+t(a_{n-1}+ta_n)))$; Horner form uses n multiplies+n adds. Weierstrass: for continuous f on closed [a,b], ∀ε>0 ∃ polynomial p with |f−p|<ε on [a,b]. Polynomial interpolation at distinct $t_0,\ldots,t_n$: Vandermonde Ax=y, $A_{ij}=t_i^j$. ***High-degree equally spaced interpolation can be ill-conditioned and oscillatory; use low-degree pieces.***

Cubic spline

Knots $a=t_0<\cdots<t_n=b$. $s(t)=s_i(t)$ on $[t_{i-1},t_i]$, degree$(s_i)\le3$, and $s\in C^2[a,b]$. Interpolation $s(t_i)=y_i$; at interior knots $s_i(t_i)=s_{i+1}(t_i),\quad s_i'(t_i)=s_{i+1}'(t_i),\quad s_i''(t_i)=s_{i+1}''(t_i)$. Two endpoint conditions required:

natural: $s''(a)=s''(b)=0$; $\quad$ clamped: $s'(a),s'(b)$; $\quad$ not−a−knot: $s_1'''(t_1)=s_2'''(t_1),\ s_{n-1}'''(t_{n-1})=s_n'''(t_{n-1})$.

Uniform $h=t_i-t_{i-1}$, $d_i=(y_i-y_{i-1})/h$, $M_i=s''(t_i)$. For $t\in[t_{i-1},t_i]$:

$s_i(t)=\frac{M_{i-1}(t_i-t)^3}{6h}+\frac{M_i(t-t_{i-1})^3}{6h}+\left(\frac{y_{i-1}}{h}-\frac{M_{i-1}h}{6}\right)(t_i-t)+\left(\frac{y_i}{h}-\frac{M_ih}{6}\right)(t-t_{i-1})$

Interior system: $\frac{h}{6}M_{i-1}+\frac{2h}{3}M_i+\frac{h}{6}M_{i+1}=d_{i+1}-d_i$, i=1,…,n−1, plus endpoint equations. Natural: $M_0=M_n=0$; coefficient matrix symmetric tridiagonal diagonally dominant PD; Thomas solve O(n).

5. CUBIC SPLINES & SPARSE MATRICES (CONT.)

Sparse structure

Sparsity=#zeros/(mn); density=nnz/(mn). Diagonal density≈1/n; tridiagonal≈3/n. Hidden sparse product: $(I+BB^T)x=x+B(B^Tx),\quad B\in R^{n\times r},\quad r\ll n$. Store/operate on nonzeros. Fill-in=new nonzeros during factorisation; PAQ=LU can reduce fill-in. For banded A=LU without disruptive pivoting, L/U retain lower/upper bandwidth; SPD $A=R^TR$ gives R same upper bandwidth.

1-D boundary-value finite difference

$-u''=f$ on $(0,1),\quad u(0)=\alpha,\quad u(1)=\beta,\quad x_j=j\Delta x,\quad \Delta x=\frac{1}{n+1}$.

$-\frac{U_{j+1}-2U_j+U_{j-1}}{(\Delta x)^2}=f(x_j)$, j=1,…,n; $U_0=\alpha$, $U_{n+1}=\beta$; error $O((\Delta x)^2)$ under sufficient smoothness. Matrix tridiagonal Toeplitz SPD with diagonal 2 and off-diagonal −1; RHS endpoints add α,β.

6. FOURIER ANALYSIS & FFT

Periodic and trigonometric form

$g(t+T)=g(t),\quad f=\frac{1}{T},\quad \cos(\omega t):T=\frac{2\pi}{\omega}$. Harmonic k: frequency k/T, angular frequency 2πk/T.

$g_n(t)=a_0+\sum_{k=1}^n\left[a_k\cos\left(\frac{2\pi kt}{T}\right)+b_k\sin\left(\frac{2\pi kt}{T}\right)\right]$.

Over any full period (shown on [−T/2,T/2]): $a_0=\frac{1}{T}\int_{-T/2}^{T/2}g(t)\,dt$;

$a_k=\frac{2}{T}\int_{-T/2}^{T/2}g(t)\cos\left(\frac{2\pi kt}{T}\right)dt$; $b_k=\frac{2}{T}\int_{-T/2}^{T/2}g(t)\sin\left(\frac{2\pi kt}{T}\right)dt$. Orthogonality: equal nonzero sine/sine or cosine/cosine integral=T/2; unequal or sine/cosine=0.

If periodic g is piecewise differentiable: Fourier series converges to g at continuity points and to [g(t+)+g(t−)]/2 at jumps. Gibbs oscillations persist near jumps.

Complex Fourier series

$g(t)=\sum_{k\in Z}c_ke^{i2\pi kt/T},\quad c_k=\frac{a_k-ib_k}{2},\quad c_{-k}=\frac{a_k+ib_k}{2}$; $c_k=\frac{1}{T}\int_{-T/2}^{T/2}g(t)e^{-i2\pi kt/T}\,dt$; $c_0=a_0$. ***Pointwise equality stated at continuity points.***

DFT / inverse / FFT

$\omega=e^{i2\pi/n}$. For j,k=0,…,n−1: $\hat{y}_k=\sum_{j=0}^{n-1}y_je^{-i2\pi jk/n}=\sum_{j=0}^{n-1}y_j\omega^{-jk}$;

$y_j=\frac{1}{n}\sum_{k=0}^{n-1}\hat{y}_ke^{i2\pi jk/n}$. Direct O(n²); FFT O(n log₂n).

$y_j\in R:\quad \hat{y}_{n-k}=\overline{\hat{y}_k},\quad \hat{y}_0\in R,\quad \hat{y}_{n/2}\in R\ (n\text{ even})$.

Fourier coefficients from equally spaced data

$t_j=j\Delta t$, Δt=T/n, $y_j\approx g(t_j)$, p=⌊n/2⌋.

$A_0=\frac{\hat{y}_0}{n},\quad A_k=\frac{2\mathrm{Re}(\hat{y}_k)}{n},\quad B_k=-\frac{2\mathrm{Im}(\hat{y}_k)}{n}$.

If n=2p even: k=1,…,p−1 use ordinary form; $A_p=\hat{y}_p/n$, $B_p=0$. If n=2p+1 odd: ordinary form through k=p. Then $y_j=A_0+\Sigma_k[A_k\cos(2\pi kt_j/T)+B_k\sin(2\pi kt_j/T)]$.

De-trending and spectrum

For $Y_j\approx x_1+x_2t_j+g(t_j)$, g periodic: least-squares fit x_1+x_2t and define $y_j=Y_j-(x_1+x_2t_j)$ before DFT. Power spectrum $p_k=\frac{|\hat{y}_k|^2}{n}$ plotted against frequency k/T; inspect k=0,…,⌊n/2⌋ for real data. Large p_k identify dominant periodic components; period=T/k (***k>0***).

7. IMPLIED VOLATILITY & NONLINEAR EQUATIONS

Black–Scholes call and implied volatility

$\tau=T-t>0$. $c = SN(d_1) - Ke^{-r\tau}N(d_2)$;

$d_1 = \frac{\ln(S/K)+r\tau}{\sigma\sqrt{\tau}} + \frac{1}{2}\sigma\sqrt{\tau}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}$; $N(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{x} e^{-\xi^2/2}\,d\xi$.

Put-call parity: $c+Ke^{-r\tau}=p+S$.

Require S>0, K>0, σ>0, τ>0; model assumptions in Topic 10.

Implied volatility: $f(\sigma) = c(\sigma) - c_{\text{mkt}} = 0, \qquad f'(\sigma) = SN'(d_1)\sqrt{\tau} > 0$.

Existence and uniqueness when $\max(S-Ke^{-r\tau},0) \le c_{\text{mkt}} \le S$.

$f(0^+) = \max(S-Ke^{-r\tau},0) - c_{\text{mkt}}, \qquad f(\infty) = S - c_{\text{mkt}}$.

Taylor and finite differences

If $f\in C^{n+1}[a,b]$, $x,x+h\in[a,b]$: $f(x+h) = \sum_{k=0}^{n} \frac{h^k}{k!} f^{(k)}(x) + O(h^{n+1})$.

$f'(x) = \frac{f(x+h)-f(x)}{h} + O(h) = \frac{f(x)-f(x-h)}{h} + O(h)$ ***Require f∈C² and shifted point in [a,b].***

$f'(x) = \frac{f(x+h)-f(x-h)}{2h} + O(h^2)$ ***Require f∈C³ and x±h∈[a,b].***

$f''(x) = \frac{f(x+h)-2f(x)+f(x-h)}{h^2} + O(h^2)$ ***Require f∈C⁴ and x±h∈[a,b].***

Machine-rounding balance:

$O(h)+O(\varepsilon/h): h \asymp \varepsilon^{1/2}; \quad O(h^2)+O(\varepsilon/h): h \asymp \varepsilon^{1/3}; \quad O(h^2)+O(\varepsilon/h^2): h \asymp \varepsilon^{1/4}$.

h must be representable so x±h differs from x.

Scalar iterations

$e^{(k)} = x^{(k)} - x^*, \qquad \beta = \lim_{k\to\infty} \frac{e^{(k+1)}}{(e^{(k)})^\nu}$; for ν=1 require |β|<1.

Newton: $x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}$. Quadratic locally; ***require f(x*)=0, f′ and f″ continuous near x*, f′≠0 there, initial guess sufficiently close***; asymptotic ratio →|f″(x*)|/(2|f′(x*)|).

Secant: $x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f[x^{(k-1)},x^{(k)}]}$; order (1+√5)/2 locally.

Require two distinct initial iterates, nonzero divided differences, starts sufficiently close to a simple root.

Systems / Gauss–Newton

$r:\mathbb{R}^n\to\mathbb{R}^m$: $J_{ij} = \frac{\partial r_i}{\partial x_j}, \qquad r(x+d) = r(x) + J(x)d + O(\|d\|^2)$. For m=n,

Newton–Raphson: $J(x^{(k)})d^{(k)} = -r(x^{(k)}), \qquad x^{(k+1)} = x^{(k)} + d^{(k)}$; ***J must be nonsingular at each solve; local quadratic convergence under smoothness and a sufficiently close start.***

For m≥n nonlinear least squares $\min\|r\|_2^2$: $J^TJd = -J^Tr$.

Local step is uniquely defined when J has full column rank.

$J_{ij} \approx \frac{r_i(x+he_j)-r_i(x)}{h}, \qquad J_{ij} \approx \frac{r_i(x+he_j)-r_i(x-he_j)}{2h}$; use $h\approx\varepsilon^{1/2}$ forward, $h\approx\varepsilon^{1/3}$ central.

8. NUMERICAL INTEGRATION

Probability integrals and quadrature

PDF p: p≥0, ∫ℝp=1. $E[f(X)] = \int_{-\infty}^{\infty} f(x)p(x)\,dx, \qquad P(x) = \int_{-\infty}^{x} p(\xi)\,d\xi$.

If P is differentiable and strictly increasing, y=P(x) gives $E[f(X)]=\int_0^1 f(P^{-1}(y))dy$.

$I(f) = \int_a^b f(x)\,dx, \qquad Q(f) = \sum_i w_i f(\xi_i), \qquad E = I - Q$. Degree of precision m: exact for all polynomials degree≤m and not all of degree m+1.

Composite rules

Grid $a=x_0<\cdots<x_N=b$, $h_i=x_i-x_{i-1}$. Trapezoidal: $Q_N^{\text{trap}} = \sum_{i=1}^{N} \frac{h_i}{2}(f_{i-1}+f_i)$. Uniform h=(b−a)/N: $h[\tfrac{1}{2}f_0+\sum_{i=1}^{N-1}f_i+\tfrac{1}{2}f_N]$.

Midpoint: $Q_N^{\text{mid}} = \sum_{i=1}^{N} h_i f\!\left(\frac{x_{i-1}+x_i}{2}\right)$.

$|E_N^{\text{trap}}| \le \frac{(b-a)Mh^2}{12}, \qquad |E_N^{\text{mid}}| \le \frac{(b-a)Mh^2}{24}$. ***Require f∈C²[a,b] for O(h²).*** Both have degree 1.

Composite Simpson, ***uniform grid and even N***:

$Q_N^{S} = \frac{h}{3}[f_0 + 4f_1 + 2f_2 + \cdots + 4f_{N-1} + f_N]$. Degree 3; $|E_N^S| \le \frac{(b-a)\max|f^{(4)}|h^4}{2880}$.

Require f∈C⁴[a,b] for O(h⁴).

$|E_{2N}| \approx \frac{|E_N|}{2^r}, \qquad r \approx \log_2\!\left(\frac{|E_N|}{|E_{2N}|}\right)$.

Gauss–Legendre

On [−1,1], N nodes/weights chosen exact through x^{2N-1}: degree $2N-1$.

$N=2:\ \xi = \pm\frac{1}{\sqrt{3}},\ w=1; \qquad N=3:\ \xi=0,\ \pm\sqrt{\frac{3}{5}},\ w=\frac{8}{9},\frac{5}{9},\frac{5}{9}$.

$\int_a^b f(x)\,dx \approx \frac{b-a}{2}\sum_{i=1}^{N} w_i f\!\left(\frac{a+b}{2}+\frac{b-a}{2}\xi_i\right)$.

Non-smooth and multidimensional

Formal orders are not guaranteed when required derivatives are missing/unbounded. Split at nonsmooth points or change variables to regularise before applying a high-order rule.

Tensor product: $\int_{[0,1]^2} f(x,y)\,dx\,dy \approx \sum_{i,j} w_i w_j f(\xi_i,\xi_j)$. A d-fold (N+1)-point rule uses $(N+1)^d$ points (**curse of dimensionality**).

9. RANDOM NUMBERS & SIMULATION — DISTRIBUTIONS

True random: physical source. Pseudorandom: deterministic, reproducible, statistically random-like. Quasirandom: deterministic, designed for multidimensional uniformity.

$X\sim U[a,b]: \quad p(x) = \frac{1}{b-a}, \quad E[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}$ (***a<b***); if U∼U[0,1], Y=a+(b−a)U.

$U\sim U[0,1], \qquad Y = F^{-1}(U) \Longrightarrow F_Y = F$. ***Use a suitable generalised inverse if strict invertibility fails.***

$Y\sim N(\mu,\sigma^2): \quad p(y) = \frac{e^{-(y-\mu)^2/(2\sigma^2)}}{\sqrt{2\pi}\sigma}, \qquad Y = \mu + \sigma Z$ (***σ>0***). Independent sums: $E\Sigma X_i=\Sigma EX_i$; $\text{Var}\Sigma X_i=\Sigma\text{Var}X_i$. CLT: iid sums are approximately normal for large n and finite variance.

9. RANDOM NUMBERS & SIMULATION — MC/QMC

Monte Carlo

$E[f] = \int_{R^d} f(x)p(x)\,dx, \qquad x_i \overset{\text{iid}}{\sim} p$

$E_N[f] = \frac{1}{N}\sum_{i=1}^{N} f(x_i)$

Estimator is unbiased; RMS error is independent of d.

$e_N[f] = \frac{\sigma(f)}{\sqrt{N}} = O(N^{-1/2})$

$e_N[f] \le \text{tol} \quad\Longrightarrow\quad N \ge \left(\frac{\sigma(f)}{\text{tol}}\right)^2$

Require independent samples and finite Var[f].

Variance reduction

Control variate: known $\mu_g=E[g]$, estimate $E[f-g]+\mu_g$; useful when Var(f−g)≪Var(f).

Antithetic pairs: ***N even***; on symmetric $\mathbb{R}^d$ pair x and −x; on $[0,1]^d$ pair x and 1−x. Odd component integrates exactly.

Importance sampling: $q(x) = g(x)p(x), \qquad E[f] \approx \frac{1}{N}\sum_{i=1}^{N}\frac{f(y_i)}{g(y_i)}, \qquad y_i \sim q$.

Require q≥0, ∫q=1, and g(x)≠0 wherever f(x)≠0; finite variance of f/g under q.

Quasi-Monte Carlo

Deterministic points:

$Q_N = \frac{1}{N}\sum_{j=1}^{N} f(x_j), \qquad x_j \in [0,1]^d$

Koksma–Hlawka:

$\left|\int_{[0,1]^d} f - Q_N\right| \le V(f)D_N^*$

Requires bounded variation V(f). Star discrepancy:

$D_N^* = \sup_{y\in[0,1]^d}\left|\prod_{i=1}^{d} y_i - \frac{\#\{x_j\in[0,y)\}}{N}\right|, \qquad D_N^* \le C\frac{(\log N)^d}{N}$

Constant C depends on d.

10. STOCHASTIC INTEGRATION & SDEs — FOUNDATIONS

Random walk / Wiener process

$t_n=n\Delta t$, $\Delta t=T/N$:

$\Delta X_n = \mu\Delta t + \sigma(\Delta t)^\gamma Z_n, \quad E[X_N - X_0] = \mu T, \quad \text{Var}(X_N - X_0) = \sigma^2 T(\Delta t)^{2\gamma-1}$;

iid $Z_n\sim N(0,1)$. Finite nonzero limit requires $\gamma=\frac{1}{2}$.

$S_{n+1} = S_n\left(1+\mu\Delta t + \sigma\sqrt{\Delta t}\,Z_n\right)$.

Wiener W: W(0)=0; a.s. continuous paths; increments over non-overlapping intervals are independent: for a<b≤c<d, W(b)−W(a) is independent of W(d)−W(c);

$W(t)-W(s)\sim N(0,t-s), \qquad W_{j+1} = W_j + \sqrt{\Delta t}\,Z_j$.

Itô integral / lemma

$\int_0^T f(W(t),t)\,dW(t) = \lim_{N\to\infty}\sum_{n=0}^{N-1} f(W(t_n),t_n)\,[W(t_{n+1})-W(t_n)]$ (***left endpoints; adapted integrand***).

$\sum_n(\Delta W_n)^2 \longrightarrow T, \qquad \int_0^T W\,dW = \frac{1}{2}W(T)^2 - \frac{1}{2}T$.

$dX = f(X,t)\,dt + g(X,t)\,dW, \qquad X(0)=X_0$. For Y=H(X,t):

$dY = \left(H_t + \frac{1}{2}g^2 H_{xx}\right)dt + H_x\,dX$. ***Require H continuously differentiable in t and twice continuously differentiable in x.***

10. STOCHASTIC INTEGRATION & BLACK–SCHOLES

Geometric Brownian motion

$dS/S=\mu dt+\sigma dW$, ***S₀>0, σ>0***. $S(t) = S_0\exp\left[\left(\mu-\frac{1}{2}\sigma^2\right)t + \sigma W(t)\right]$;

$\ln\!\left(\frac{S(t)}{S_0}\right) \sim N\!\left(\left(\mu-\frac{1}{2}\sigma^2\right)t, \sigma^2 t\right)$; S(t) is lognormal and positive.

Black–Scholes PDE and conditions

$V_t + rSV_S + \frac{1}{2}\sigma^2S^2V_{SS} = rV$, S>0, 0<t<T. Delta hedge uses $f=V_S$ short shares. ***Assumptions: GBM underlying; constant known r,σ; no dividends, transaction costs, taxes or arbitrage; continuous trading and short selling allowed.***

$c(S,T)=\max(S-K,0),\ c(0,t)=0,\ c\sim S-Ke^{-r(T-t)}; \qquad p(S,T)=\max(K-S,0),\ p(0,t)=Ke^{-r(T-t)},\ p\to 0$.

Put-call parity: $S+p=c+Ke^{-r(T-t)}$ (***European options, same K,T***).

Log-price/time transformation

y=ln(S/K), τ=T−t, V(S,t)=U(y,τ):

$-U_\tau + \left(r-\frac{1}{2}\sigma^2\right)U_y + \frac{1}{2}\sigma^2U_{yy} = rU$; y∈ℝ, τ∈[0,T].

$U(y,0) = K\max(e^y-1,0), \quad U\to 0\ (y\to-\infty), \quad U\sim Ke^y - Ke^{-r\tau}\ (y\to\infty)$;

$U(y,0) = K\max(1-e^y,0), \quad U\sim Ke^{-r\tau}\ (y\to-\infty), \quad U\to 0\ (y\to\infty)$.

11. FDMs FOR BLACK–SCHOLES

Grid, notation and differences

Truncate $S\in[0,S_{max}]$, ***S_{max}>K and sufficiently large***.

$\Delta S = \frac{S_{\max}}{N_S+1}, \quad S_j = j\Delta S; \qquad \Delta t = \frac{T}{N_t}, \quad t_n = n\Delta t, \quad V_j^n \approx V(S_j,t_n)$.

$V_t(S_j,t_n) + rS_jV_S(S_j,t_n) + \frac{1}{2}\sigma^2S_j^2V_{SS}(S_j,t_n) = rV_j^n$.

$V_t(S_j,t_n) \approx \frac{V_j^n - V_j^{n-1}}{\Delta t} \qquad$ backward, $O(\Delta t)$.

$V_S(S_j,t_n) \approx \frac{V_{j+1}^n - V_{j-1}^n}{2\Delta S}, \qquad V_{SS}(S_j,t_n) \approx \frac{V_{j+1}^n - 2V_j^n + V_{j-1}^n}{(\Delta S)^2}$; each O((ΔS)²).

Boundary/terminal values from Topic 10.

Backward Euler - EXPLICIT scheme

$\rho = \frac{\Delta t}{2(\Delta S)^2}, \quad \alpha_j = (\sigma^2S_j^2 - rS_j\Delta S)\rho, \quad \beta_j = 1 - r\Delta t - 2\sigma^2S_j^2\rho, \quad \gamma_j = (\sigma^2S_j^2 + rS_j\Delta S)\rho$.

$V_j^{n-1} = \alpha_jV_{j-1}^n + \beta_jV_j^n + \gamma_jV_{j+1}^n$. Accuracy O(Δt+(ΔS)²) when stable.

Sufficient stability: $\sigma^2 \ge r, \qquad \frac{\Delta t}{(\Delta S)^2} \le \min_{1\le j\le N_S}\frac{1}{r(\Delta S)^2 + \sigma^2S_j^2}$; then α,β,γ≥0 and |α|+|β|+|γ|≤1. Under this mesh relation overall order is O((ΔS)²).

Forward Euler - IMPLICIT scheme

$\alpha_j = \frac{j\Delta t}{2}(r-\sigma^2 j), \quad \beta_j = 1 + r\Delta t + \frac{\sigma^2S_j^2\Delta t}{(\Delta S)^2}, \quad \gamma_j = -\frac{\sigma^2S_j^2\Delta t}{2(\Delta S)^2} - \frac{rS_j\Delta t}{2\Delta S}$. Solve tridiagonal $\alpha_jV_{j-1}^n + \beta_jV_j^n + \gamma_jV_{j+1}^n = V_j^{n+1}$ with boundary terms in RHS, backwards $n=N_t-1,\ldots,0$. Accuracy O(Δt+(ΔS)²); unconditionally stable. ***σ²≥r is sufficient for strict diagonal dominance/nonsingularity, not necessary.*** Factor once if r,σ time-independent; total $O(N_tN_S)$.